I&EC research
Industrial & Engineering Chemistry Research
pubs.acs.org/IECR
Article

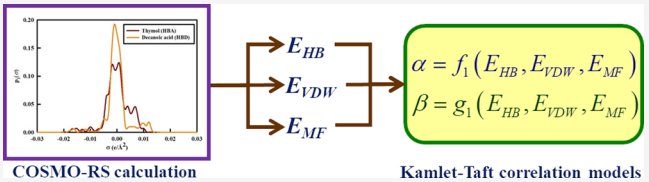
First-Principles Prediction of Kamlet–Taft Solvatochromic Parameters of Deep Eutectic Solvent Using the COSMO-RS Model

Debashis Kundu, Priyansh Singh Rao, and Tamal Banerjee*

Cite This: *Ind. Eng. Chem. Res.* 2020, 59, 11329–11339
Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Rapid discovery of a deep eutectic solvent (DES) and its increasing application in various scientific applications demands a well-defined polarity scale for the new generation of eutectic solvents. The tunable nature of a DES facilitates to increase its number by combining various hydrogen bond acceptors and donors in multiple molar ratios. Therefore, the experimental measurement of polarity scale represented by solvatochromic parameters is a cumbersome task. In this context, we have developed a first-principle-based conductor-like screening model-real solvent (COSMO-RS)-derived predictive modeling approach to predict the Kamlet–Taft solvatochromic parameters of 26 hydrophilic and 62 hydrophobic DESs. The COSMO-RS-derived energies, that is, hydrogen bonding energy (E_{HB}), misfit energy (E_{MF}), and van der Waals energy (E_{VDW}), are linearly correlated with the experimental data set of 28 DESs having 4 choline chloride, 17 tetraalkylammonium organic salt, 4 menthol, and 3 betaine subclasses. A lowest root-mean-square deviation (rmsd) of 0.06% for the choline chloride class of DESs and a highest rmsd of 3.14% for the tetrabutylammonium chloride class of DESs are obtained. Therefore, the correlated models are then extended to develop 60 DESs consisting of 15 choline chloride DESs, 4 different hydrophilic DESs, 17 tetraalkylammonium organic salt origin DESs, 7 menthol DESs, and 17 thymol-based DESs.



1. INTRODUCTION

A deep eutectic solvent (DES) is rapidly emerging as a new kind of green solvent prepared from low-cost abundant resources. It is prepared by a mixture of organic salt and hydrogen bond donor (HBD) in a definitive molar ratio with the melting point lower than that of the individual components. Therefore, the formation of a stable DES depends on the strength of the HBD and hydrogen bond acceptor (HBA). Often, the organic salt contains a cation and an anion, where the cation acts as an additional HBD. However, a natural compound such as menthol or thymol can form a DES with carboxylic acid. Here, menthol or thymol acts as a HBA and carboxylic acid acts as a HBD. By varying the molar ratio of the components, combining different organic salts with HBD, or mixing natural compounds and carboxylic acids, a large number of DESs has been proposed in the literature.[1,2] Therefore, the physiochemical properties of DESs can be tuned depending on the molar compositions of the respective components.[3] The tunable properties open up the feasibility of application of DES as a solvent in phase equilibria,[4] catalysis,[5] and chemical synthesis.[6] The combination of HBA and HBD gives DES the ability to accept and donate protons during solvation and thus form a hydrogen bond with the solute. This enhances the dissolution capability of DES, due to which it is widely used as a solvent in separation and purification processes.[7,8]

One of the important properties of a solvent is polarity. Polarity defines the possible interactions between a solute and a solvent. Additionally, the polarity of a solvent influences the reaction rate, mechanism, solvation ability, and product yield.[9] The experimental measurement of polarity involves the analysis of shifting UV–vis absorbance spectra of solvatochromic probes. A simplistic polarity scale is a one-parameter Dimroth–Reichardt $E_T(30)$ scale which can only provide qualitative trends.[10] The hydrogen bond donating ability of a solvent influences the $E_T(30)$ polarity scale, which merely measures the polarizability. Kamlet and Taft[11] have developed a multiparametric model using a set of solvatochromic probes to assess the ability of HBD (acidity) and HBA (basicity) and polarizability of the same solvent. The Kamlet–Taft equation in the form of linear solvation energy relationship is given by

$$XYZ = (XYZ)_0 + s(\pi^* + d\delta) + a\alpha + b\beta \quad (1)$$

Here, XYZ correlates the energy property of solvent parameters, whereas the solvent-independent coefficients are denoted by a, b, d, and s. The correlation of energy property of solvent parameters of the reference system is denoted by $(XYZ)_0$. Further, α denotes the hydrogen bond (H-bond) donating ability or acidity, β denotes the hydrogen bond accepting ability or basicity, π^* represents the polarizability of

Received: February 4, 2020
Revised: May 5, 2020
Accepted: May 21, 2020
Published: May 21, 2020



Downloaded via WESTLAKE UNIV on October 22, 2025 at 09:34:48 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

ACS Publications
© 2020 American Chemical Society

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329–11339

dipolarity of the solvent, and δ is a correction term. 4-Nitroaniline, *N*,*N*-diethyl-4-nitroaniline, and 2,6-dichloro-4-(2,4,6-triphenyl-*n*-pyridino)-phenolate better known as Reichardt's betaine dye 33 are traditionally being used for the experimental measurement of K−T parameters. However, all the dyes are expensive. Further, the selection of dye and experimental approach influence the solvatochromic parameters.[12] Therefore, the continuous improvement of experimental data set coupled with the expensive experiment makes it harder to follow a definitive chart of polarity scale.

As a state-of-the-art green solvent and its ever-growing application in the separation and purification processes, construction of polarity chart becomes utmost necessity. However, a handful of literature studies are available regarding the experimental measurement of K−T parameters of DESs. Six ChCl-based DESs, four menthol-based DESs, and three quaternary ammonium-based DESs are reported by Florindo et al.[13] The K-T parameters of 13 ChCl-based DESs and 3 betaine-based DESs are being reported by Cai et al.[14] Teles et al. further reported the α and β values of 19 quaternary ammonium salt-based DESs.[15] The K−T parameters of two quaternary ammonium salt-based DESs and one betaine-based DES are reported by Dietz et al.[16] However, like other solvents, the reported data of solvatochromic parameters may vary upon the change of dye.[15] The HBD and HBA present in the DES make distinctive interactions with the solute. Further, the inherent H-bond interaction between the HBA and HBD increases the complexity of nonbonded interactions with the solute. Because of the large number of DES, it will be an expensive endeavor for the experimental measurement of solvatochromic parameters of every DES. Therefore, a priori modeling of solvatochromic parameters and construction of polarity scale would ensure benchmark database for the future applications of DESs. However, to date, there is no modeling approach for the solvatochromic parameters of DESs. Hence, a modeling technique with a similar class of compounds having analogous physiochemical behavior and applications would create a benchmark step for future development.

The combinatorial nature of construction, tunable nature in physiochemical properties, and their similar applications bring DESs in close analogy with ionic liquids (ILs). Therefore, to construct a predictive polarity chart of DES, we can highlight some of the modeling approaches of the solvatochromic parameters reported for ILs. To establish a well-defined, consistent, and predictive approach of solvatochromic parameters, several modeling approaches have been attempted in the literature for the case of ILs. Mente and Maroncelli performed the Monte Carlo (MC) simulation of solvent interactions in the $E_T(30)$ scale of solvent polarity.[17] Znamenskiy and Kobrak explored the polarity of ILs using a solvatochromic dye by molecular dynamics (MD) simulation.[18] However, both the methods are computationally expensive. Contreras et al. proposed an empirical energy density model based on the Lewis molecular acidity of ILs.[19] Niedermeyer et al. proposed time-dependent density functional theory (TD-DFT) calculation involving the interaction between solvents and K−T dyes for the prediction of K−T parameters.[20] Recently, Cláudio et al.[21] and Kurnia et al.[22] predicted β and α values using the quantum chemical (QC)-based conductor-like screening model-real solvent (COSMO-RS) approach. The unimolecular QC calculation of cation and anion and subsequent correlation with the β and α values shall open up a computationally inexpensive rapid screening of solvents.

In this context, we have developed a QC-based predictive approach for the hydrogen bond (H-bond) acidity (α) and hydrogen bond basicity (β) of DESs. First-principle-based predictive model COSMO-RS has been used to predict the various nonbonded interactions of DESs. The predicted nonbonded interactions are then correlated with the experimentally measured values of α and β. The correlation coefficients are later employed to develop an extended scale of predictive models.

2. COMPUTATIONAL METHOD

2.1. COSMO-RS Theory. COSMO-RS is a QC-based predictive model, where various thermodynamic properties are calculated depending on the structure of molecules. Here, molecules are assumed to be submerged in an ideal virtual conductor environment. The interactions are thereby calculated at the interface of the molecules and conductor, considering the counter-polarization and electrostatic screening. Thereafter, the molecules are considered as a collection of surface segments, and the screening charge of each segment is computed. The screening charge density, position, and area of each segment of the surface are stored in a file named "COSMO" file (.cosmo extension). Later, the pairwise interactions between the two segments are computed with a set of statistical mechanical calculations given in the literature.[23,24] Misfit energy originates from the nonideal contact of two segments. van der Waals' energy is a summation of dispersion energy and cavity formation energy. The cavity formation energy arises because of the formation of a cavity inside the conductor medium to accept a molecule from the gas phase to the conductor medium. The inhomogeneous charges on the molecular surface are adequately represented by the dispersion energy. Hydrogen bond (H-bond) plays a crucial role in the formation of DES. Therefore, we have used the modified version of COSMO-RS to adequately represent the H-bond interaction in the predictive modeling.[25]

The screening charge densities of surface segments are standardized to accommodate into equally spaced 71 discrete bins from −0.035 to +0.035 e/Å^2 (an increment of 0.001 e/Å^2). Therefore, a normalized histogram of segmental areas is plotted within the limits of bins, better known as sigma profile (σ-profile). The σ-profile is the characteristic distribution of charge densities and is unique for every molecule. However, modified COSMO-RS theory segregates the hydrogen bonding segments using a Gaussian-type probability in the original σ-profile. The same is not added with the nonhydrogen bonding segmental areas. The segregation results in the representation of two σ-profiles, that is, the hydrogen bonding σ-profile and the nonhydrogen bonding σ-profile. Considering the possible attraction among the atomic pairs, the dispersion energy term is calculated by the first-order mean field term. Further, the cavity formation energy is calculated from the thermodynamics perturbation theory. The description of each energetic term and the corresponding COSMO parameters are adapted from the literature.[25,26]

2.2. COSMO File Generation. A two-step approach has been considered for the generation of COSMO file. In the first step, the geometries of cations, anions, and HBDs are drawn using the GaussView 5.0[27] visualization package. The geometries of the isolated species are optimized using the Gaussian 09 QC package using DFT.[28] Becke's three-

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

parameter-based exact exchange functional coupled with the gradient correlation of Lee, Young, and Parr, better known as the B3LYP level of theory, is employed to optimize the geometry in accordance with the 6-311+G(d) basis set.[29,30] In the COSMO calculation step, the BVP86 basis set is used to further optimize the geometries.[31] The later optimization is carried out in combination with the density fitting basis sets DGA1 and SVP[32,33] to improve the relative energies and molecular structure. The newly generated COSMO file (.cosmo extension) is used in our in-house reimplementation of COSMO code to predict the respective interaction energies.[34]

2.3. Construction of Models. Figure 1 shows the flowchart of model construction and subsequent extended scale of prediction of K−T parameters. The geometries of the HBAs and HBDs are independently optimized by QC calculation, followed by the generation of COSMO file. After the generation of COSMO file, the in-house reimplementation of COSMO-RS calculation gives three specific interaction energies, namely, hydrogen bonding energy (E_{HB}), misfit energy (E_{MF}), and van der Waals energy (E_{VDW}).[34] Therefore, a priori predictive models of H-bond acidity and basicity are linearly correlated with the three specific interactions. The linearized forms are given as

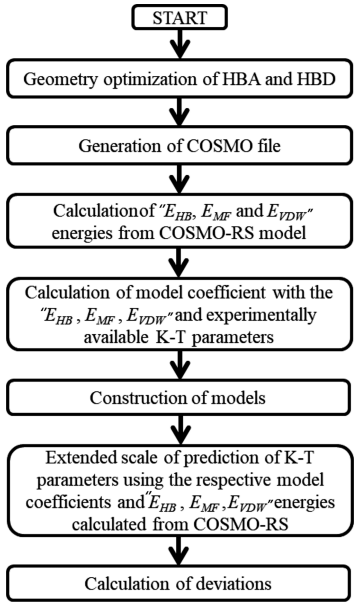


Figure 1. Flowchart of model development and extended scale prediction.

$$\alpha = A_\alpha \frac{E_{HB}}{\sum_n E_{HB}} + B_\alpha \frac{E_{MF}}{\sum_n E_{MF}} + C_\alpha \frac{E_{VDW}}{\sum_n E_{VDW}} + D_\alpha \quad (2)$$

$$\beta = A_\beta \frac{E_{HB}}{\sum_n E_{HB}} + B_\beta \frac{E_{MF}}{\sum_n E_{MF}} + C_\beta \frac{E_{vdw}}{\sum_n E_{vdw}} + D_\beta \quad (3)$$

Here, A_α, B_α, C_α, and D_α and A_β, B_β, C_β, and D_β are two sets of coefficients used in eqs 2 and 3, respectively. However, the order of each of the interaction energies is different. Hence, to accommodate the importance of all the three energies, the calculated energies are normalized over their respective subclasses. The classification of subclasses of DESs will be discussed in detail in Section 3.2. Therefore, the coefficients of eqs 2 and 3 are calculated by fitting the interaction energies of a particular subclass of DES with the experimental values of K−T parameters. However, to calculate the coefficients of a particular subclass of DES, all known experimental K−T parameters of that subclass are considered. Computed coefficients are then used to predict the α and β values of individual compounds across the subclass. For the extended scale prediction, the corresponding coefficients of subclasses are taken along with the respective energies of that compound to predict the α and β values. By this way, the K−T parameters of the new DES are determined, whose experimental values are not available. The deviation of the predicted K−T parameters with the experimental values are determined by rmsd (%) and is given by

$$\text{rmsd} = 100 \times \left[\sum_{i=1}^{m} \frac{([\text{K} - \text{T}]_i^{\text{pred}} - [\text{K} - \text{T}]_i^{\text{expt}})^2}{m} \right]^{1/2} \quad (4)$$

Here, [K−T] defines the α or β parameter, "pred" defines the model-predicted values, "expt" defines the experimental values, and m represents the total number of DESs in a subclass. However, the absolute deviation is given by

$$\text{absolute deviation (\%)} = 100 \times |[\text{K} - \text{T}]_i^{\text{pred}} - [\text{K} - \text{T}]_i^{\text{expt}}| \quad (5)$$

3. RESULTS AND DISCUSSION

The COSMO-RS-based predictive modeling has been used on 88 types of DESs, of which 26 are of hydrophilic class and the rest belong to the hydrophobic class of DESs. The hydrophilic class is further subdivided into 19 choline chloride (ChCl)-based DESs, 3 other choline cation-based DESs, 3 betaine class of DESs, and 1 ethyl(2-hydroxyethyl) dimethylammonium class of DES. The hydrophobic class is further subdivided to 34 tetraalkylammonium salt-based DESs (including chloride and bromide anions), 11 menthol-based DESs, and 17 thymol-based DESs. The complete number of DESs along with the model prediction and the extended scale of the predicted K−T parameters are given in Table S1 of the Supporting Information. Further, the structure of HBAs and HBDs and their respective energies E_{HB}, E_{MF}, and E_{VDW} are given in Tables S2−S6 of the Supporting Information.

3.1. Development of Models. The magnitude of predicted energies can vary with the variation of COSMO-RS model parameters, that is, (a_{eff}, c_{hb}). With the variation in predicted energies, the magnitude of predicted K−T parameters will also vary. Therefore, we have used three sets of {a_{eff}, c_{hb}} (given in Table S7 of the Supporting Information), and subsequently, three sets of H-bond basicity of ChCl-based DESs are predicted. Figures S1−S3 of the Supporting Information represent the correlational plots of β parameter, and from these correlation plots, the {a_{eff} = 7.50 Å^2, c_{hb} = 85580 kcal/mol Å^4/e^2} set is selected as the optimum model parameter for subsequent predictions. The details of COSMO-RS parameters are given in Tables S8−S9 of the Supporting Information. In addition to the predicted energies, various combinations of two-parameter, three-parameter, and four-parameter models are constructed for every subclass of DES. The predicted model coefficients and correlation coefficient (R^2) of each combination of every subclass are given in Tables S10−S25 of the Supporting Information. Out of the various

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

Table 1. System Equations for α_{pred} along with the Constants Fitted for Various Classes of DESs

HBA[a]	A_α	B_α	C_α	D_α	R^2	adjR^2	model name
ChCl	−8.4053	0	67.3805	−13.3013	0.9966	0.9899	M1α
$[N_{4444}]$Br	52.5078	−36.2109	623.5207	−105.6360	0.9840	0.9734	M2α
$[N_{4444}]$Cl	0	0	−77.9368	16.4714	0.8579	0.8104	M3α
DL-Menthol	−3.9993	3.1385	0	1.9052	0.9434	0.8302	M4α
betaine	2.0997	0	0	0.7041	0.9831	0.9662	M5α

[a]Name and abbreviation of DES: choline chloride: ChCl; tetrabutylammonium chloride: $[N_{4444}]$Cl; tetrabutylammonium bromide: $[N_{4444}]$Br. $\text{adj}R^2 = 1 - \left(\frac{(1-R^2)(n-1)}{n-k-1}\right)$, where n is the number of data points and k is the number of independent regressors.

Table 2. System Equations for β_{pred} along with the Constants Fitted for Various Classes of DESs

HBA	A_β	B_β	C_β	D_β	R^2	adjR^2	model name
ChCl	−15.6079	2.0535	0	3.2269	0.9578	0.9196	M1β
$[N_{4444}]$Br	5.3977	0	372.9387	−62.1327	0.9045	0.8408	M2β
$[N_{4444}]$Cl	-0.1946×10^3	0.1533×10^3	-2.4671×10^3	0.5028×10^3	0.9994	0.9977	M3β
DL-menthol	0	1.5001	−35.0697	8.9549	0.9645	0.8935	M4β
Betaine	0	0	−407.737	136.457	0.9491	0.8981	M5β

Table 3. Experimental and Predicted Values of α and β for Different HBA Systems as per the Classifications Given in Tables 1 and 2

name of DES (HBA/HBD)[a]	molar ratio (HBA/HBD)	α_{exp}	α_{pred}	absolute deviation (%)	rmsd (%)	β_{exp}	β_{pred}	absolute deviation (%)	rmsd (%)	references	predication model used
ChCl/MalA	1:2	1.39	1.3912	0.12	0.23	0.42	0.4197	0.03	0.06	13	M1α, M1β
ChCl/Urea	1:2	1.42	1.4176	0.24		0.50	0.5000	0.00			
ChCl/Ethy	1:2	1.47	1.4732	0.32		0.57	0.5694	0.06			
ChCl/Gly	1:2	1.49	1.4879	0.21		0.52	0.5209	0.09			
$[N_{4444}]$Br/ButA	1:2	1.02	1.0259	0.59	0.64	0.81	0.8282	1.82	3.14	15	M2α, M2β
$[N_{4444}]$Br/ButA	1:1	1.09	1.0912	0.12		0.84	0.8185	2.15			
$[N_{4444}]$Br/ButA	2:1	0.94	0.9393	0.07		0.82	0.8293	0.93			
$[N_{4444}]$Br/HexA	1:2	1.02	1.0110	0.90		0.93	0.9506	2.06			
$[N_{4444}]$Br/OctA	1:2	0.98	0.9735	0.65		1.09	1.0245	6.55			
$[N_{4444}]$Br/DecA	1:2	0.95	0.9592	0.92		1.05	1.0888	3.88			
$[N_{4444}]$Cl/ButA	1:2	0.92	0.9247	0.47	0.08	0.99	0.9891	0.09	0.27		M3α, M3β
$[N_{4444}]$Cl/HexA	1:2	0.90	0.8849	1.51		1.02	1.0195	0.05			
$[N_{4444}]$Cl/OctA	1:2	0.84	0.8609	2.09		1.19	1.1949	0.49			
$[N_{4444}]$Cl/DecA	1:2	0.85	0.8416	0.84		1.28	1.2765	0.35			
$[N_{4444}]$Cl/DecA	1:1	0.91	0.9079	0.21		1.21	1.2101	0.01			
DL-menthol/AceA	1:1	1.64	1.6093	3.07	2.25	0.60	0.5946	0.54	0.71	13	M4α, M4β
DL-menthol/LevA	1:1	1.56	1.5876	2.76		0.58	0.5906	1.06			
DL-menthol/OctA	1:1	1.77	1.7840	1.40		0.50	0.5021	0.21			
DL-menthol/DodA	2:1	1.79	1.7792	1.08		0.57	0.5627	0.73			
Bet/Gly	1:2	1.253	1.2840	3.10	0.18	0.587	0.4550	13.2	2.68	14	M5α, M5β
Bet/CitA	1:1	1.671	1.6691	0.19		0.012	0.096	8.4			
Bet/Urea	1:2	1.288	1.2589	2.91		1.035	1.083	4.8			

[a]Name and abbreviation of DES: choline chloride malonic acid: ChCl/MalA; choline chloride urea: ChCl/urea; choline chloride ethylene glycol: ChCl/Ethy; choline chloride glycerol: ChCl/Gly; tetrabutylammonium bromide butyric acid: $[N_{4444}]$Br/ButA; tetrabutylammonium bromide hexanoic acid: $[N_{4444}]$Br/HexA; tetrabutylammonium bromide octanoic acid: $[N_{4444}]$Br/OctA; tetrabutylammonium bromide decanoic acid: $[N_{4444}]$Br/DecA; tetrabutylammonium chloride butyric acid: $[N_{4444}]$Cl/ButA; tetrabutylammonium chloride hexanoic acid: $[N_{4444}]$Cl/HexA; tetrabutylammonium chloride octanoic acid: $[N_{4444}]$Cl/OctA; tetrabutylammonium chloride decanoic acid: $[N_{4444}]$Cl/DecA; DL-menthol acetic acid: DL-menthol/AceA; DL-menthol levulinic acid: DL-menthol/LevA; DL-menthol octanoic acid: DL-menthol/OctA; DL-menthol dodecanoic acid: DL-menthol/DodA; betaine glycerol: Bet/Gly; betaine citric acid: Bet/CitA; and betaine urea: Bet/urea.

combinations, only those models which have the highest R^2 value within a particular subclass are selected for the benchmarking of models.

3.2. Benchmarking of Models. The benchmarking prediction of α and β with respect to the experimental data set are performed either by the classification of HBAs or HBDs. We had to adopt two separate sets of classifications based on the experimental measurement of α and β with respect to the difference in dyes. The experimental database considered in this work has taken Reichardt's betaine dye 33,[13,14] 4-nitoaniline,[13−15] *N*,*N*-diethyl-4-nitroaniline,[13−15] and pyridine-*N*-oxide.[15] It should be noted that the acidic nature of DES interferes with the solvatochromic nature of dye.[13] Hence, for the DES, the derivative of Reichardt's betaine dye 30, that is, Reichardt's betaine dye 33, is used to avoid the zwitterionic nature of the probe. However, in none of the literature studies, the purity of DES as well as the water content is explicitly mentioned. However, as the polarity parameters depend on the

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

Table 4. System Equations for α_{pred} along with the Constants Fitted for Various Classes of DESs

HBD[a]	A_α	B_α	C_α	D_α	R^2	adjR^2	model name
ButA	0	0	−269.4030	90.7510	0.9874	0.9748	M6α
HexA	0	0	−275.7970	92.8589	0.9348	0.8697	M7α
OctA	0	−0.5640	0	1.0880	0.9878	0.9756	M8α

[a]Name and abbreviation of DESs: butanoic acid: ButA; hexanoic acid: HexA; and octanoic acid: OctA.

interaction of solvent and dye, variation of water content will affect the measurement. The developed model coefficients and the corresponding model numbers to predict ChCl, DL-menthol, [N_{4444}]Cl, [N_{4444}]Br, and betaine are given in Tables 1 and 2. Further, the respective models are developed based on four ChCl, four DL-menthol, five [N_{4444}]Cl, six [N_{4444}]Br, and three betaine-based DESs. The corresponding HBDs of these DESs are given in Tables 3 and 4. The subclasses are formed according to the HBA of DES because HBAs often play a dominant role in the overall polarity of DES.[13] In all of the cases, the four-parameter linearized models are used. However, it is observed from Tables 1 and 2 that the four-parameter model reduces to three- or two-parameter models depending on the subclasses of DESs. The correlation coefficients are obtained by the fitting of experimental data set with the COSMO-RS-derived energies. Each of the interaction terms is normalized with respect to the energies of the compounds of the same subclass to reduce the dominance of any interaction energy over others. Despite that, the linearized fitting often results in a null value for the one or two coefficients and reduces the four-parameter models into two- or three-parameter models. Therefore, the null values of the model coefficients of E_{HB}, E_{MF}, and E_{VDW} are only of quantitative significance. The model coefficients of each subclass are obtained using the calculation procedure given in Figure 1. The correlative plots of α and β are further visualized in Figure 2i–iv. Very good values of correlation coefficients (R^2) are observed for the prediction of α except for the class of [N_{4444}]Cl. However, R^2 values are higher for the prediction of β (Table 2). Further, considering the diverse chemical nature of DES, separate models for each subclass are formed with respect to the experimental data set. The subclass-oriented model construction produces higher accuracy of prediction, as observed from the correlation coefficient values. Table 3 represents the predicted H-bond acidity and basicity values of each compound of ChCl, DL-menthol, [N_{4444}]Cl, [N_{4444}]Br, and betaine subclasses, which are considered for the model development in Tables 1 and 2. It is observed that the rmsd of predicted H-bond acidity lies within 0.08–2.25%, while for H-bond basicity, it is within 0.06–3.14%. The lower values of rmsd imply the accurate prediction of the developed models.

However, in many cases, keeping the same HBA, new DESs are formed by increasing the alkyl chain length of the HBD. This is a common phenomenon for the tetraalkylammonium-based DESs, which can have large amount of possible DES by increasing the alkyl chain length of HBD. To construct the model of K–T parameters, we have grouped the tetraalkylammonium HBAs as per their HBDs. The common HBDs of butanoic acid (ButA) of tetraethylammonium, tetrapropylammonium, and tetrabutylammonium chloride are grouped into one subclass so that the effect of chain length on the K–T parameters can be addressed. The same is said about the hexanoic acid (HexA)- and octanoic acid (OctA)-based DESs in Tables 4 and 5. The effect of chain length on the K–T parameters will be useful for the extended scale prediction of

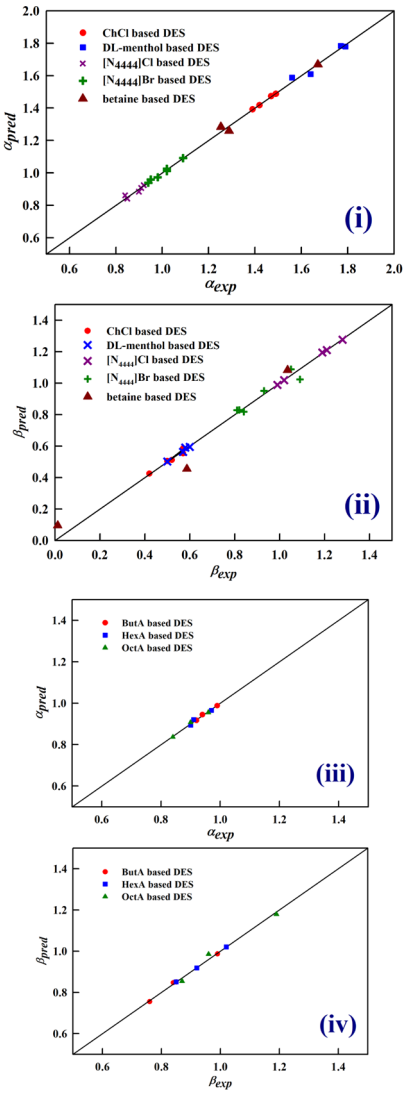


Figure 2. (i) Correlation among experimental (α_{exp}) and predicted (α_{pred}) values of hydrogen bond acidity for different HBA systems, (ii) correlation among experimental (α_{exp}) and predicted (α_{pred}) values of hydrogen bond acidity for different HBD systems, (iii) correlation among experimental (β_{exp}) and predicted (β_{pred}) values of hydrogen bond basicity for different HBA systems, and (iv) correlation among experimental (β_{exp}) and predicted (β_{pred}) values of hydrogen bond basicity for different HBD systems.

higher-order tetraalkylammonium-based DES. Therefore, a different set of correlative models is developed based on the alkyl chain length of HBDs, given in Tables 4 and 5 against the experimental data set of Teles et al.[15] The corresponding HBAs of these DESs are given in Tables 6 and 7. For both α and β, we observe that the two-parameter models yield a higher correlation coefficient. Further, the predicted α and β, given in Table 6, have lower range of rmsds, that is, 0.33–0.79% for α and 0.08–1.84% for β. The lower rmsds indicate

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329–11339

Table 5. System Equations for β_{pred} along with the Constants Fitted for Various Classes of DESs

HBD	A_β	B_β	C_β	D_β	R^2	adjR^2	model name
ButA	0.9253	0	0	0.5549	0.9970	0.9940	M6β
HexA	0	0.7396	0	0.6835	0.9998	0.9996	M7β
OctA	1.5570	0	0	0.4877	0.9813	0.9626	M8β

Table 6. Experimental and Predicted Values of α and β for Different HBD Systems as per the Classifications Given in Tables 4 and 5

name of DES (HBA/HBD)[a]	molar ratio (HBA/HBD)	α_{exp}	α_{pred}	absolute deviation (%)	rmsd (%)	β_{exp}	β_{pred}	absolute deviation (%)	rmsd (%)	references	predication model used
$[N_{2222}]$Cl/ButA	1:2	0.99	0.9882	0.18	0.33	0.76	0.7556	0.44	0.52	15	M6α, M6β
$[N_{3333}]$Cl/ButA	1:2	0.94	0.9446	0.46		0.84	0.8473	0.73			
$[N_{4444}]$Cl/ButA	1:2	0.92	0.9172	0.28		0.99	0.9871	0.29			
$[N_{2222}]$Cl/HexA	1:2	0.97	0.9657	0.43	0.79	0.85	0.8507	0.07	0.08		M7α, M7β
$[N_{3333}]$Cl/HexA	1:2	0.91	0.9211	1.11		0.92	0.9188	0.12			
$[N_{4444}]$Cl/HexA	1:2	0.90	0.8932	0.68		1.02	1.0205	0.05			
$[N_{2222}]$Cl/OctA	1:2	0.96	0.9555	0.45	0.54	0.87	0.8546	1.54	1.84		M8α, M8β
$[N_{3333}]$Cl/OctA	1:2	0.90	0.9075	0.75		0.96	0.9859	2.59			
$[N_{4444}]$Cl/OctA	1:2	0.84	0.8370	0.30		1.19	1.1795	1.05			

[a]Name and abbreviation of DES: tetraethylammonium chloride butyric acid: $[N_{2222}]$Cl/ButA; tetrapropylammonium chloride butyric acid: $[N_{3333}]$Cl/ButA; tetrabutylammonium chloride butyric acid: $[N_{4444}]$Cl/ButA; tetraethylammonium chloride hexanoic acid: $[N_{2222}]$Cl/HexA; tetrapropylammonium chloride hexanoic acid: $[N_{3333}]$Cl/HexA; tetrabutylammonium chloride hexanoic acid: $[N_{4444}]$Cl/HexA; tetraethylammonium chloride octanoic acid: $[N_{2222}]$Cl/OctA; tetrapropylammonium chloride octanoic acid: $[N_{3333}]$Cl/OctA; and tetrabutylammonium chloride octanoic acid: $[N_{4444}]$Cl/OctA.

that the choice of classification plays an essential role for the accurate prediction of K−T parameters.

3.3. Extended Scale of Prediction. The α parameter describes the capacity of a solvent to donate protons in solvent to solute H-bonding, whereas the β parameter describes the ability of accepting a proton.[15] The predictive models (M1α-M8α and M1β-M8β) are hereby used to predict 19 types of hydrophilic and 41 types of hydrophobic DESs. To the best of our knowledge, there is no experimental data available for these DESs. Models M1α and M1β are used to predict 15 types of choline chloride, 3 types of choline-based, and 1 type of ethyl(2-hydroxyethyl) dimethylammonium chloride based hydrophilic DESs. The predicted K−T parameters are given in Table 7. Despite the same HBA, that is, ChCl, the predicted α and β parameters of choline-based DESs do not follow any specific trend. Similarly, the predicted α and β parameters of other choline-based DESs do not follow any trend because of the wide variation of structure of HBD. Other than the chloride-containing choline-based DES, we have predicted the α and β parameters of fluoride, tetrafluoroborate, and nitrate anion containing choline cations in HBA salt and urea as the HBD. The stronger cation−anion interaction is due to the fluoride anion and nitrate anion, where the predicted acidity is lower than that of the ChCl/urea DES. However, the β parameter relates the proton accepting ability and largely depends on the system environment. Therefore, a clear trend cannot be established here.

Models M2α-M4α and M2β-M4β are used for the extended scale of prediction of 11 types of tetrabutylammonium-based, 7 types of DL-menthol-based, and 17 types of thymol-based hydrophobic DESs and are given in Table 8. Unlike the hydrophilic class of DES, the hydrophobic DES mostly follows a specific trend for α and β parameters. The increasing alkyl chain length of HBDs increases the hydrophobicity. For the $[N_{4444}]$Br-based DES, an increment of α and β is observed because of the increment in the alkyl chain length of OctA, DecA, and LauA. $[N_{4444}]$Cl/LauA produces a similar trend of α and β. The anion in HBA salt plays a major role in the overall acidity of DES. The chloride anion displays a stronger cation−anion interaction over the bromide anion and therefore has greater ability to accept the proton. Therefore, the chloride-based DESs have lower proton donating ability for solvent−solute interaction. Hence, they possess lower acidity than bromide-based and tetraalkylammonium-based DESs. However, the β value must follow the opposite trend of α value. This increasing trend of β value is followed in the set of {$[N_{4444}]$Br/AceA; $[N_{4444}]$Br/AcrA; $[N_{4444}]$Br/ProA} and {$[N_{4444}]$Br/OctA; $[N_{4444}]$Br/DecA; $[N_{4444}]$Br/LauA; $[N_{4444}]$Br/OleA}. However, a stronger cation−anion interaction might be an obstacle in order to get a clear trend on the β value of $[N_{4444}]$Cl-based DESs. The α values of DL-menthol/lauA decrease with the increment of a molar ratio of HBA. On the contrary, the β values follow a reverse trend. The opposite trend of α and β values conclude the fact that the DL-menthol-based DESs follow the polarity nature of tetraalkylammonium-based DESs. Further, the alkyl chain length of HBD has minimal effect on the overall polarity of DL-menthol-based DESs compared to the molar ratio of HBA. A general decreasing trend of α and β values are obtained with the decrease of HBA molar ratio in thymol-based DESs. Looking at the chemical structure of thymol (Table S5 of the Supporting Information), the −OH moiety of thymol is capable of donating proton during solvation. However, with the increasing amount of HBD in the respective series of thymol DES, the molar ratio of HBA decreases, resulting in the decrease of H-bond acidity parameter. Despite the increment in the molar ratio of HBD for the thymol-based DES, the basicity parameter also decreases. This is a contradictory behavior compared with the tetraalkylammonium salt-based DES. However, in the tetraalkylammonium-based DES, the anion of HBA salt plays a major role in the reverse trend of β against α. However, with the absence of salt form of HBA, the β parameter gives a similar trend like α. Therefore, it can be concluded that the HBDs do not play a central role in the

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

Table 7. Extended Values of α and β for Different Hydrophilic DES Systems[a]

name of DES (HBA/HBD)[b]	molar ratio (HBA/HBD)	α_{pred}	β_{pred}	model used
ChCl/thiourea	1:2	1.4111	0.4977	M1α, M1β
ChCl/1-methyl urea	1:2	1.6028	0.7397	
ChCl/1,3-dimethyl urea	1:2	1.6958	0.8572	
ChCl/1,1-dimethyl urea	1:2	1.7393	0.9126	
ChCl/acetamide	1:2	1.7891	1.0613	
ChCl/benzamide	1:2	1.6824	0.8118	
ChCl/2,2,2-Tfa	2:5	2.1233	1.5706	
ChCl/imidazole	3:7	1.8148	1.1014	
ChCl/BenA	1:1	1.3706	0.5046	
ChCl/OxaA	1:1	1.1921	0.2707	
ChCl/PheA	1:1	1.1254	0.1065	
ChCl/LevA	1:2	1.5529	0.6102	
ChCl/D-IsoA	1:2	1.3835	0.3219	
ChCl/T-CinA	1:1	1.146	0.1384	
ChCl/resorcinol	1:4	1.8506	1.0029	
ChF/urea	1:2	1.3066	0.3628	
$ChBF_4$/urea	1:2	1.4131	0.3505	
$ChNO_3$/urea	1:2	1.3838	0.3156	
ethyl(2-hydroxyethyl) dimethylammonium chloride/urea	1:2	1.1259	0.0682	

[a]The full names of DESs are given at the bottom of the table. [b]Name and abbreviation of DES: choline chloride thiourea: ChCl/thiourea; choline chloride 1-methyl urea: ChCl/1-methyl urea; choline chloride 1,3-dimethyl urea: ChCl/1,3-dimethyl urea; choline chloride 1,1-dimethyl urea: ChCl/1,1-dimethyl urea; choline chloride acetamide: ChCl/acetamide; choline chloride benzamide: ChCl/benzamide; choline chloride 2,2,2-trifluoroacetamide: ChCl/2,2,2-Tfa; choline chloride imidazole: ChCl/imidazole; choline chloride benzoic acid: ChCl/BenA; choline chloride oxalic acid: ChCl/OxaA; choline chloride phenylacetic acid: ChCl/PheA; choline chloride levulinic acid: ChCl/LevA; choline chloride D-isosorbide acid: ChCl/D-IsoA; choline chloride *trans*-cinnamic acid: ChCl/T-CinA; choline chloride resorcinol: ChCl/resorcinol; choline fluoride urea: ChF/urea; choline tetrafluoroborate urea: $ChBF_4$/urea; choline nitrate urea: $ChNO_3$/urea; and ethyl(2-hydroxyethyl)dimethylammonium chloride urea: ethyl(2-hydroxyethyl)dimethylammonium chloride/urea.

Table 8. Extended Values of α and β for Different Hydrophobic DES Systems

name of DES (HBA/HBD)[a]	molar ratio (HBA/HBD)	α_{pred}	β_{pred}	model used
$[N_{4444}]$Br/AceA	1:2	1.2386	0.6395	M2α, M2β
$[N_{4444}]$Br/ProA	1:2	1.0694	0.7425	
$[N_{4444}]$Br/AcrA	1:2	1.0341	0.6471	
$[N_{4444}]$Br/OctA	1:1	1.1597	1.0000	
$[N_{4444}]$Br/DecA	1:1	1.1779	1.0613	
$[N_{4444}]$Br/LauA	1:1	1.2408	1.1306	
$[N_{4444}]$Br/OleA	1:2	1.3755	1.3469	
$[N_{4444}]$Cl/AceA	1:1	1.0241	0.8170	M3α, M3β
$[N_{4444}]$Cl/OctA	1:1	0.9220	1.1633	
$[N_{4444}]$Cl/LauA	1:1	0.8962	1.0377	
$[N_{4444}]$Cl/LauA	1:2	0.8262	1.1312	
DL-menthol/LacA	1:1	1.5796	0.6535	M4α, M4β
DL-menthol/LacA	1:2	1.8519	0.4463	
DL-menthol/PyrA	1:2	1.6088	0.6017	
DL-menthol/LauA	1:1	1.8299	0.4899	
DL-menthol/LauA	3:2	1.8073	0.5286	
DL-menthol/LauA	2:1	1.7844	0.5635	
DL-menthol/LauA	5:2	1.7619	0.5950	
thymol/UndA	7:3	1.8840	0.3628	
thymol/UndA	3:2	1.8787	0.3375	
thymol/UndA	1:1	1.8759	0.3196	
thymol/UndA	1:2	1.8736	0.2988	
thymol/UndA	1:3	1.8730	0.2908	
thymol/UndA	1:4	1.8728	0.2866	
thymol/DecA	3:2	1.8924	0.3313	
thymol/DecA	1:1	1.8897	0.3137	
thymol/DecA	1:2	1.8875	0.2933	
thymol/DecA	1:3	1.8869	0.2856	
thymol/camphor	7:3	1.8907	0.3033	
thymol/camphor	3:2	1.8849	0.2818	
thymol/camphor	1:1	1.8816	0.2665	
thymol/DL-menthol	1:1	1.8424	0.5478	
thymol/DL-menthol	1:2	1.8353	0.4900	
thymol/lidocaine	2:1	1.8552	0.3615	
thymol/lidocaine	1:1	1.8515	0.3310	

[a]Name and abbreviation of DES: tetrabutylammonium chloride acetic acid: $[N_{4444}]$Cl/AceA; tetrabutylammonium chloride octanoic acid: $[N_{4444}]$Cl/OctA; tetrabutylammonium chloride lauric acid: $[N_{4444}]$Cl/LauA; tetrabutylammonium bromide acetic acid: $[N_{4444}]$Br/AceA; tetrabutylammonium bromide propionic acid: $[N_{4444}]$Br/ProA; tetrabutylammonium bromide octanoic acid: $[N_{4444}]$Br/OctA; tetrabutylammonium bromide decanoic acid: $[N_{4444}]$Br/DecA; tetrabutylammonium bromide acrylic acid: $[N_{4444}]$Br/AcrA; tetrabutylammonium bromide lauric acid: $[N_{4444}]$Br/LauA; tetrabutylammonium bromide oleic acid: $[N_{4444}]$Br/OleA; DL-menthol lactic acid: DL-menthol/LacA; DL-menthol pyruvic acid: DL-menthol/PyrA; DL-menthol lauric acid: DL-menthol/LauA; thymol undecylenic acid: thymol/UndA; thymol decanoic acid: thymol/DecA; thymol camphor: thymol/camphor; thymol DL-menthol: thymol/DL-menthol; and thymol lidocaine: thymol/lidocaine.

polarity of thymol-based DES. However, for all the thymol-based DESs, we have used M4α and M4β models because among the available experimental α and β data sets, M4α and M4β are developed on the basis of ring structure.

Models M6α-M8α and M6β-M8β are developed by varying the alkyl chain length of HBD as well as varying the HBAs from tetraethylammonium to tetrabutylammonium moieties (Table 6). The increment of alkyl chain length of HBAs with respect to fixed HBD facilitates the extended scale of prediction of 3 tetraheptylammonium and 3 tetraoctylammonium DESs. It is observed from Table 9 that for a fixed HBD (say ButA), α decreases, whereas β increases with the increased chain length of HBA. Further, for a fixed HBD (say tetraheptylammonium), the α parameter decreases with the increasing alkyl chain length of HBD. However, with respect to tetrabutylammonium-based DES, tetraheptylammonium- and tetraoctylammonium-based DESs have lower acidity values. This is related to the increase in alkyl chain length, where the target species finds difficulty to accept or donate protons, resulting in lower interaction strength and thus lower acidity. For the extended scale prediction of β parameter, the tetraoctylammonium DESs have higher β values than the corresponding lower alkyl chain-based tetraalkylammonium-based DESs. In this case, the argument of decreasing trend of β values with the increment is established.

Therefore, the above-derived predicted data set of α and β never attains very high values. All the predicted values are in

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

Table 9. Extended Values of α and β for Different Hydrophobic DES Systems[a]

name of DES (HBA/HBD)[a]	molar ratio (HBA/HBD)	α_{pred}	β_{pred}	model used
$[N_{7777}]$Cl/ButA	1:2	0.7658	1.5024	M6α, M6β
$[N_{8888}]$Cl/ButA	1:2	0.6941	1.7521	
$[N_{7777}]$Cl/HexA	1:2	0.7498	1.3860	M7α, M7β
$[N_{8888}]$Cl/HexA	1:2	0.6821	1.5606	
$[N_{7777}]$Cl/OctA	1:2	0.5624	1.9711	M8α, M8β
$[N_{8888}]$Cl/OctA	1:2	0.4319	2.3402	

[a]Name and abbreviation of DES: tetraheptylammonium chloride butyric acid: $[N_{7777}]$Cl/ButA; tetraoctylammonium chloride butyric acid: $[N_{8888}]$Cl/ButA; tetraheptylammonium chloride hexanoic acid: $[N_{7777}]$Cl/HexA; tetraoctylammonium chloride hexanoic acid: $[N_{8888}]$Cl/HexA; tetraheptylammonium chloride octanoic acid: $[N_{7777}]$Cl/OctA; and tetraoctylammonium chloride octanoic acid: $[N_{8888}]$Cl/OctA.

similar range to that reported for the limited experimental α and β data sets of DES. Further, the predicted data set is within the similar range of ILs.[21,22] Therefore, the reliability of extended scale of prediction can be safely concluded and treated as polarity scale for H-bond acidity and H-bond basicity. Because of the combinatorial nature of DES, an enormous number of DESs can be theorized, and many of them can actually be synthesized in the laboratory. Herein, we have taken a careful screening approach of dividing DESs into various subclasses, and accordingly, models are being formed (M1α-M8α and M1β-M8β). Further, the extended scale of prediction is restricted to that particular subclass only. This careful screening leads to the enhanced accuracy of prediction. However, there are a few DESs whose experimental K−T parameters are reported. Therefore, we encourage the scientific community that, whenever at least three K−T parameters are determined experimentally, the K−T parameters of other compounds belonging to that subclass can be predicted reliably with the approach developed in this manuscript.

3.4. Comparative Analysis with the Predictive Models in the Literature. Table 10 represents the comparative analysis of predictive approaches of determining K−T parameters described in the literature with the developed method given in this work. Znamenskiy and Kobrak[18] applied MD simulation to determine the polarity of IL. Herein, the solute−solvent interaction was replicated between the interactions of donor moiety of imidazolium IL with the acceptor moiety of betaine-30 solvent through MD simulation.

Table 10. Comparison of Modeling Approaches of K−T Parameters of DES with the Literature

article	predictive method and compounds	key outcome	references
Znamenskiy and Kobrak	MD simulation of 1-butyl-3-methylimidazolium hexafluorophosphate and betaine-30	solute−solvent interaction was elaborated between the interactions of donor moiety of imidazolium IL with the acceptor moiety of betaine-30	18
		nonspecific electrostatic interactions are key contributor to the polarity	
		localized electrostatic screening leads to the origin of nonspecific interactions	
Mente and Maroncelli	MC simulation of 12 solvents and betaine-30	solvent polarity with betaine-30 dye for $E_T(30)$ scale	17
Contreras et al.	empirical energy density models from DFT on 45 imidazolium-based ILs including 12 Bronsted acid ILs	excess electronic chemical potential descriptor model which determines the electron accepting power of IL	19
		the second model is the correlation of local charge capacity of the cation in its valence state with the Lewis molecular acidity of the IL	
Niedermeyer et al.	TD-DFT of various classes of cations and anions of IL	TD-DFT calculations for interactions among the Kamlet−Taft dyes and IL	20
		computation of HOMO−LUMO at optical excitation to determine α parameter from the complex Reichardt's dye with cation of IL	
		similar calculation of HOMO−LUMO with 4-nitroaniline-anion of IL for the β parameter	
Cláudio et al.	COSMO-RS model on ILs having 1-butyl-3-methylimidzolium cations	estimated H-bond interaction energies in IL	21
		linear correlations of experimental H-bond basicity values with the H-bond interaction energies	
		extended level of polarity scale derived from the correlations	
Kurnia et al.	COSMO-RS model on ILs having bis(trifluoromethyl sulfonyl)imide anion	experimental determination of K−T parameters of IL	22
		correlation of K−T parameters with the COSMO-RS-derived H-bond, vdW, misfit energies	
		extended scale of model prediction from the correlational model	
present study	QC calculation for the optimization of HBA and HBD structures, followed by COSMO-RS model calculation on 88 DES (26 hydrophilic and 62 hydrophobic)	classification of DES as per the HBA and HBD	
		correlation of experimental K−T parameters with the COSMO-RS-derived H-bond, vdW, misfit energies	
		construction of predictive K−T parameter models	
		extended scale of prediction is based on the correlational K−T parameter models	

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

Further, the authors identified the nonspecific electrostatic interactions, which are key contributors to the polarity, arising from the localized electrostatic screening. However, the localized nature of nonspecific interactions only applies in the thermodynamic limit. Mente and Maroncelli[17] performed MC simulation of 12 solvents with betaine −30 dye to explore the solvent polarity at molecular basis of the $E_T(30)$ scale. The optimized electronic structures and charge distributions are calculated from ab initio and semiempirical calculations. The combination of solute and solvent interaction employed in the MD and MC simulations are more specific to the solvatochromism study of a particular dye. Contreras et al.[19] presented two empirical energy density models based on the DFT of 45 ILs. The first model is based on the excess electronic chemical potential descriptor, which determines the electron accepting power of IL. The second approach modeled the local charge capacity of the cation in its valence state. The Kamlet−Taft α parameter is correlated with the combination of these two models. The electronic structure-based calculations opened up an alternative way of classifying H-bond acidity. Niedermeyer et al.[20] employed TD-DFT calculations to calculate the interactions among the Kamlet−Taft dyes and ILs to predict its K−T parameters. This technique opens up the possibility of predicting the solvatochromic properties of ILs, which are not discovered yet. TD-DFT-formulated computation of highest occupied molecular orbital (HOMO)−lowest unoccupied molecular orbital (LUMO) at optical excitation determines the α parameter from the complex Reichardt's dye with a cation of IL, whereas a similar calculation with the 4-nitroaniline-anion of IL predicts the β parameter. Cláudio et al.[21] and Kurnia et al.[22] developed a framework based on the COSMO-RS model to predict the α and β parameters of IL. H-bond, vdW, and misfit energies derived from the COSMO-RS calculation are fitted into empirical models for the prediction of α and β parameters. However, the authors only predicted the α of IL having bis(trifluoromethyl sulfonyl)imide anion and β parameter of 1-butyl-3-methylimidzolium cations. With this work, we claim as the first group to report the predictive data set of α and β of K−T polarity scale of DES. The a priori prediction relies on the correlation of E_{HB}, E_{MF}, and E_{VDW} energies derived from the COSMO-RS calculation. The DESs are classified into various subclasses according to their HBA and HBD structures and extended scale of data set is constructed from the respective models of the subclass. The subclass-oriented modeling improves the accuracy of predictions with respect to the experimental data set.

3.5. Suitable Methodologies of Prediction of K−T Parameters and Future Scope of Development. In the previous section, predictive approaches such as MD−MC simulation, QC calculations, and COSMO-RS framework are discussed with respect to the possible modeling pathways of K−T parameters. In the MD and MC approaches, the K−T parameters were predicted with respect to the interactions with the solvatochromic dye. However, force fields of ILs as well as DES are not reliably developed. Therefore, the accuracy of the predicted data set always remains questionable when adopting the MD approach. However, both the approaches are computationally expensive, and considering the ever-growing number of DES, the construction of such database with these approaches will take a long time. Therefore, these approaches are not suitable for a priori prediction. On the other hand, the QC-based calculation of the DES−dye complex will be relatively faster. However, generalized models based on the electronic structure calculation will likely accumulate large error in the database. This work is also derived from the optimized geometry from the electronic structure calculation coupled with the COSMO-RS formulation. In order to reduce the large error, it has been observed from the derived data set that an individual model based on the subclasses of DES shall improve the accuracy. However, the accuracy of our prediction mainly depends on the accurate theoretical description with respect to various interaction energies in the COSMO-RS framework. Therefore, the predicted and extended scale of database can be further improved by incorporating various interaction parameters given in the literature.[35,36] The existing experimental methodology of K−T parameters involves extensive solvatochromic probe. For an ever-growing list of DES, the continuous experimentation poses serious economic challenge. Further, the experimental polarity scale depends on the set of solvatochromic dyes as well as purity of the DES. Hence, the present methodology emphasizes on the identification of subclasses of DES, followed by the creation of experimental data set of small number of DES of that subclass. Next, the predictive models developed within the framework of this work can predict the extended scale of K−T parameters of other DESs in the same subclass. However, a futuristic development could be the generalized model based on the group interaction parameter. However, this depends on the accurate description of interaction energies. Based on the accurate description of interaction energies of functional moieties of DES, the interaction parameters can be listed and from that the list of generalized model can be formulated.

4. CONCLUSIONS

COSMO-RS-predicted hydrogen bonding, misfit, and van der Waals energies were calculated and correlated in a four-parameter model to predict the Kamlet−Taft H-bond acidity and H-bond basicity of DESs. Altogether, α and β parameters were predicted of 88 DES comprising hydrophilic and hydrophobic type. The available experimental data set of 7 hydrophilic and 21 hydrophobic DESs was used to construct 8 H-bond acidity and 8 H-bond basicity models. The constructed models predicted α and β values with deviation less than 2.25 and 3.14%, respectively. Therefore, these models were used for the extended scale of prediction of 19 hydrophilic and 41 hydrophobic DESs comprising choline, tetraalkylammonium, DL-menthol, and thymol-based DESs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.iecr.0c00574.

Structure of HBAs, HBDs, COSMO-RS parameters, COSMO-RS-predicted energies, benchmarking models, and plots (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Tamal Banerjee − *Department of Chemical Engineering, Indian Institute of Technology Guwahati, Guwahati, Assam 781039, India;* orcid.org/0000-0001-8624-6586; Phone: +91 361 2582266; Email: tamalb@iitg.ac.in; Fax: +91 361 2582291

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339

Authors

Debashis Kundu − *Department of Chemical Engineering, Indian Institute of Technology Guwahati, Guwahati, Assam 781039, India*

Priyansh Singh Rao − *Department of Chemical and Biochemical Engineering, Indian Institute of Technology Patna, Patna, Bihar 801106, India*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.iecr.0c00574

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors acknowledge the computational resource of Param-Ishan Supercomputing facility of Indian Institute of Technology Guwahati for the QC calculations.

■ REFERENCES

(1) Zhang, Q.; De Oliveira Vigier, K.; Royer, S.; Jérôme, F. Deep eutectic solvents: syntheses, properties and applications. *Chem. Soc. Rev.* **2012**, *41*, 7108−7146.

(2) Florindo, C.; Branco, L. C.; Marrucho, I. M. Quest for green-solvent design: From hydrophilic to hydrophobic (deep) eutectic solvents. *ChemSusChem* **2019**, *12*, 1549−1559.

(3) Smith, E. L.; Abbott, A. P.; Ryder, K. S. Deep eutectic solvents (dess) and their applications. *Chem. Rev.* **2014**, *114*, 11060−11082.

(4) Verma, R.; Mohan, M.; Goud, V. V.; Banerjee, T. Operational strategies and comprehensive evaluation of menthol based deep eutectic solvent for the extraction of lower alcohols from aqueous media. *ACS Sustainable Chem. Eng.* **2018**, *6*, 16920−16932.

(5) Phadtare, S. B.; Shankarling, G. S. Halogenation reactions in biodegradable solvent: Efficient bromination of substituted 1-aminoanthra-9,10-quinone in deep eutectic solvent (choline chloride:urea). *Green Chem.* **2010**, *12*, 458−462.

(6) Boston, R.; Foeller, P. Y.; Sinclair, D. C.; Reaney, I. M. Synthesis of barium titanate using deep eutectic solvents. *Inorg. Chem.* **2017**, *56*, 542−547.

(7) Paiva, A.; Craveiro, R.; Aroso, I.; Martins, M.; Reis, R. L.; Duarte, A. R. C. Natural deep eutectic solvents − solvents for the 21st century. *ACS Sustainable Chem. Eng.* **2014**, *2*, 1063−1071.

(8) Li, X.; Row, K. H. Development of deep eutectic solvents applied in extraction and separation. *J. Sep. Science* **2016**, *39*, 3505−3520.

(9) MacFarlane, D. R.; Pringle, J. M.; Johansson, K. M.; Forsyth, S. A.; Forsyth, M. Lewis base ionic liquids. *Chem. Commun.* **2006**, 1905−1917.

(10) Spange, S.; Lungwitz, R.; Schade, A. Correlation of molecular structure and polarity of ionic liquids. *J. Mol. Liq.* **2014**, *192*, 137−143.

(11) Kamlet, M. J.; Taft, R. W. The solvatochromic comparison method. I. The .beta.-scale of solvent hydrogen-bond acceptor (HBA) basicities. *J. Am. Chem. Soc.* **1976**, *98*, 377−383.

(12) Jessop, P. G.; Jessop, D. A.; Fu, D.; Phan, L. Solvatochromic parameters for solvents of interest in green chemistry. *Green Chem.* **2012**, *14*, 1245−1259.

(13) Florindo, C.; McIntosh, A. J. S.; Welton, T.; Branco, L. C.; Marrucho, I. M. A closer look into deep eutectic solvents: exploring intermolecular interactions using solvatochromic probes. *Phys. Chem. Chem. Phys.* **2018**, *20*, 206−213.

(14) Cai, C.; Yu, W.; Wang, C.; Liu, L.; Li, F.; Tan, Z. Green extraction of cannabidiol from industrial hemp (Cannabis sativa L.) using deep eutectic solvents coupled with further enrichment and recovery by macroporous resin. *J. Mol. Liq.* **2019**, *287*, 110957.

(15) Teles, A. R. R.; Capela, E. V.; Carmo, R. S.; Coutinho, J. A. P.; Silvestre, A. J. D.; Freire, M. G. Solvatochromic parameters of deep eutectic solvents formed by ammonium-based salts and carboxylic acids. *Fluid Phase Equilib.* **2017**, *448*, 15−21.

(16) Dietz, C. H. J. T.; Kroon, M. C.; van Sint Annaland, M.; Gallucci, F. Thermophysical properties and solubility of different sugar-derived molecules in deep eutectic solvents. *J. Chem. Eng. Data* **2017**, *62*, 3633−3641.

(17) Mente, S. R.; Maroncelli, M. Computer simulations of the solvatochromism of betaine-30. *J. Phys. Chem. B* **1999**, *103*, 7704−7719.

(18) Znamenskiy, V.; Kobrak, M. N. Molecular dynamics study of polarity in room-temperature ionic liquids. *J. Phys. Chem. B* **2004**, *108*, 1072−1079.

(19) Contreras, R.; Aizman, A.; Tapia, R. A.; Cerda-Monje, A. Lewis molecular acidity of ionic liquids from empirical energy−density models. *J. Phys. Chem. B* **2013**, *117*, 1911−1920.

(20) Niedermeyer, H.; Ashworth, C.; Brandt, A.; Welton, T.; Hunt, P. A. A step towards the a priori design of ionic liquids. *Phys. Chem. Chem. Phys.* **2013**, *15*, 11566.

(21) Cláudio, A. F. M.; Swift, L.; Hallett, J. P.; Welton, T.; Coutinho, J. A. P.; Freire, M. G. Extended scale for the hydrogen-bond basicity of ionic liquids. *Phys. Chem. Chem. Phys.* **2014**, *16*, 6593−6601.

(22) Kurnia, K. A.; Lima, F.; Cláudio, A. F. M.; Coutinho, J. A. P.; Freire, M. G. Hydrogen-bond acidity of ionic liquids: an extended scale. *Phys. Chem. Chem. Phys.* **2015**, *17*, 18980−18990.

(23) Kundu, D.; Banerjee, T. Multicomponent vapor−liquid−liquid equilibrium prediction using an a priori segment based model. *Ind. Eng. Chem. Res.* **2011**, *50*, 14090−14096.

(24) Klamt, A. Conductor-Like Screening Model for Real Solvents-A new approach to the quantitative calculation of solvation phenomena. *J. Phys. Chem.* **1995**, *99*, 2224−2235.

(25) Lin, S.-T.; Chang, J.; Wang, S.; Goddard, W. A.; Sandler, S. I. Prediction of vapor pressures and enthalpies of vaporization using a COSMO solvation model. *J. Phys. Chem. A* **2004**, *108*, 7429−7439.

(26) Wang, S.; Sandler, S. I.; Chen, C.-C. Refinement of COSMO-SAC and the applications. *Ind. Eng. Chem. Res.* **2007**, *46*, 7275−7288.

(27) Dennington, R.; Keith, T.; Millam, J. *GaussView*; Semichem Inc.: Shawnee Mission, KS, 2009.

(28) Frisch, M.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Mennucci, B.; Petersson, G. A.; Nakatsuji, H.; Caricato, M.; Li, X.; Hratchian, H. P.; Izmaylov, A. F.; Bloino, J.; Zheng, G.; Sonnenberg, J. L.; Hada, M.; Ehara, M.; Toyota, K.; Fukuda, R.; Hasegawa, J.; Ishida, M.; Nakajima, T.; Honda, Y.; Kitao, O.; Nakai, H.; Vreven, M. T.; Peralta, J. E.; Ogliaro, F.; Bearpark, M.; Heyd, J. J.; Brothers, E.; Kudin, K. N.; Staroverov, V. N.; Keith, T.; Kobayashi, R.; Normand, J.; Raghavachari, K.; Rendell, A.; Burant, J. C.; Iyengar, S. S.; Tomasi, J.; Cossi, M.; Rega, N.; Millam, J. M.; Klene, M.; Knox, J. E.; Cross, J. B.; Bakken, V.; Adamo, C.; Jaramillo, J.; Gomperts, R.; Stratmann, R. E.; Yazyev, O.; Austin, A. J.; Cammi, R.; Pomelli, C.; Ochterski, J. W.; Martin, R. L.; Morokuma, K.; Zakrzewski, V. G.; Voth, G. A.; Salvador, P.; Dannenberg, J. J.; Dapprich, S.; Daniels, A. D.; Farkas, O.; Foresman, J. B.; Ortiz, J. V.; Cioslowski, J.; Fox, D. J. *Gaussian 09*, Revision B. 01; Gaussian, Inc.: Wallingford, CT.

(29) Becke, A. D. Density-Functional Thermochemistry. III. The Role of exact exchange. *J. Chem. Phys.* **1993**, *98*, 5648−5652.

(30) Lee, C.; Yang, W.; Parr, R. G. Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1988**, *37*, 785−789.

(31) Perdew, J. P. Density-Functional Approximation for the correlation energy of the inhomogeneous electron gas. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1986**, *33*, 8822−8824.

(32) Sosa, C.; Andzelm, J.; Elkin, B. C.; Wimmer, E.; Dobbs, K. D.; Dixon, D. A. a local density functional study of the structure and vibrational frequencies of molecular transition-metal compounds. *J. Phys. Chem.* **1992**, *96*, 6630−6636.

(33) Schäfer, A.; Horn, H.; Ahlrichs, R. fully optimized contracted gaussian basis sets for atoms Li to Kr. *J. Chem. Phys.* **1992**, *97*, 2571−2577.

(34) Bharti, A.; Kundu, D.; Rabari, D.; Banerjee, T. *Phase Equilibria in Ionic Liquid Facilitated Liquid−liquid Extractions*; CRC Press: Boca Raton, 2017; Vol. *1*, pp 1−222.

(35) Toure, O.; Audonnet, F.; Lebert, A.; Dussap, C.-G. COSMO-RS-PDHS: A new predictive model for aqueous electrolytes solutions. *Chem. Eng. Res. Des.* **2014**, *92*, 2873−2883.
(36) Müller, S.; González de Castilla, A.; Taeschler, C.; Klein, A.; Smirnova, I. Evaluation and refinement of the novel predictive electrolyte model COSMO-RS-ES based on solid-liquid equilibria of salts and Gibbs free energies of transfer of ions. *Fluid Phase Equilib.* **2019**, *483*, 165−174.

https://dx.doi.org/10.1021/acs.iecr.0c00574
Ind. Eng. Chem. Res. 2020, 59, 11329−11339